Name: ______________________ Date: __________

Yr 8 Mathematics (General & Ext)

Investigation 3, 2011

Topic – Measurement- In Class Validation

34
= ____ %

Total Time: *35 minutes*

Weighting: *5% of the year. (Take Home & Validation Included)*

Equipment: *Take home component, Calculator*

Penalties (only deducted once in the whole assessment):

- ❑ *Rounding* - Incorrect/inappropriate (-1)
- ❑ *Units* - Incorrect/missing (-1)

Important Information:
This investigation will be completed under test conditions. Answers should be rounded appropriately. All working should be shown in the space provided. Solutions without working may not be awarded full marks. Please take the marks for each question into account when answering the question.

AIM:
In this assessment you will be investigating area and perimeter.

Use the following information to complete the questions:

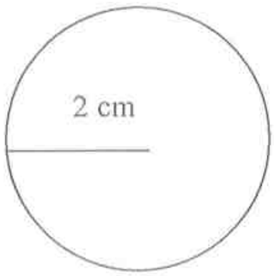
Circumference (Perimeter) of a circle = $2 \times \pi \times r$ **Area of a circle =** $\pi \times r^2$

Where r is the radius

For this investigation let $\pi = 3.14$

4. **[17 marks: 1, 2, 2, 2, 2, 3, 3, 2]**
Answer the following questions **to complete the table**:

a) A circle has a radius of 2cm. What is the circumference of this circle? Write your answer in the table.



C = 12.56cm ✓

b) Enlarge circle 1 with a scale factor of 2, ie. double the radius of circle 1.

i) What is the circumference of circle 2?

C= 25.12cm ✓

ii) How has the circumference of circle 2 changed with respect to circle 1?

It has doubled OR Enlarged by a scale factor of 2 ✓

c) Enlarge circle 1 with a scale factor of 3.

i) What is the circumference of circle 3?

C = 37. cm ✓

ii) How has the circumference of circle 3 changed with respect to circle 1?

It has tripled

OR

Enlarged by a scale factor of 3. ✓

d) Enlarge circle 1 with a scale factor of 4.

i) What is the circumference of circle 4?

C = 50.24 cm ✓

ii) How has the circumference of circle 4 changed with respect to circle 1?

It's 4 times bigger

OR

Enlarged by a scale factor of 4 ✓

e) Enlarge circle 1 with a scale factor of 5.

i) What is the circumference of circle 5?

C = 62.8 cm ✓

ii) How has the circumference of circle 5 changed with respect to circle 1?

5 times bigger

OR

Enlarged by a scale factor of 5 ✓

f) Enlarge circle 1 with a scale factor of n.

i) What is the circumference of circle n?

$C = 4\pi n$ OR $12.56n$ ✓✓

ii) How has the circumference of circle n changed with respect to circle 1?

n times bigger

OR

Enlarged by a scale factor of n ✓

g) Complete the table.

Circle Number	Radius (cm)	Circumference (cm)	Scale Factor
1	2	12.56	1
2	4	25.12	2
3	6	37.68	3
4	8	50.24	4
5	10	62.8	5
n	2n	$4\pi n$ OR $12.56n$	n

Radius ✓

Circumference ✓

Scale factor ✓

$-\frac{1}{2}$ each error

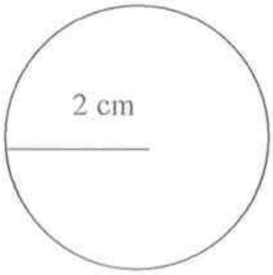
h) Does the pattern that you discovered in question 1 (Take Home Component) apply to the circumference of a circle? If yes, what is the pattern?

Yes ✓

Circumference is increased by the scale factor. ✓

4. **[17 marks: 1, 2, 2, 2, 2, 3, 3, 2]**
Answer the following questions **to complete the table**:

a) A circle has a radius of 2cm. What is the area of this circle? Write your answer in the table.



$A = 12.56 \text{ cm}^2$ ✓

b) Enlarge circle 1 with a scale factor of 2.

i) What is the area of circle 2?

$A = 50.24 \text{ cm}^2$ ✓

ii) How has the area of circle 2 changed with respect to circle 1?

It's 4 times bigger ✓

c) Enlarge circle 1 with a scale factor of 3.

i) What is the area of circle 3?

$A = 113.04 \text{ cm}^2$ ✓

ii) How has the area of circle 3 changed with respect to circle 1?

It's 9 times bigger ✓

d) Enlarge circle 1 with a scale factor of 4.

i) What is the area of circle 4?

$A = 200.96 \text{ cm}^2$ ✓

ii) How has the area of circle 4 changed with respect to circle 1?

16 times bigger ✓

e) Enlarge circle 1 with a scale factor of 5.

i) What is the area of circle 5?

314 cm^2 ✓

ii) How has the area of circle 5 changed with respect to circle 1?

25 times bigger ✓

f) Enlarge circle 1 with a scale factor of n.

i) What is the area of circle n?

$A = 4n^2\pi$ OR $12.56n^2$ ✓✓

ii) How has the area of circle n changed with respect to circle 1?

n^2 times bigger ✓

g) Complete the table.

Circle Number	Radius (cm)	Area (cm^2)	Scale Factor
1	2	12.56	1
2	4	50.24	2
3	6	113.04	3
4	8	200.96	4
5	10	314	5
n	2n	$4n^2\pi$ OR $12.56\,n^2$	n

Radius ✓
Area ✓
Scale Factor ✓
$-\frac{1}{2}$ each error

h) Does the pattern that you discovered in question 2 apply to the area of a circle? If yes, what is the pattern?

Yes ✓

Area increases by the scale factor squared ✓